

Xavier (Glorot) and He Initialisation

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1. Overview

Key point: Draw the starting weights at random with a spread set by the number of inputs to each node: variance $1/\text{fan-in}$ (Xavier, for tanh and sigmoid) or $2/\text{fan-in}$ (He, for ReLU). The signal then keeps roughly the same size through every layer.

The [weight initialisation Note](#) showed what fails. Zeros and constants make every node identical, so the weights must be random. Random weights that are too small make the signal die out; too large, and it saturates or explodes. So the spread must be in between.

This Note gives the two standard answers:

- **Xavier initialisation**, also called **Glorot initialisation** (Xavier Glorot is one person), for tanh and sigmoid;
- **He initialisation**, after Kaiming He, for ReLU and its variants.

Each comes in a **normal** version (weights from a normal distribution) and a **uniform** version (weights from a uniform distribution), and Keras has all four built in.

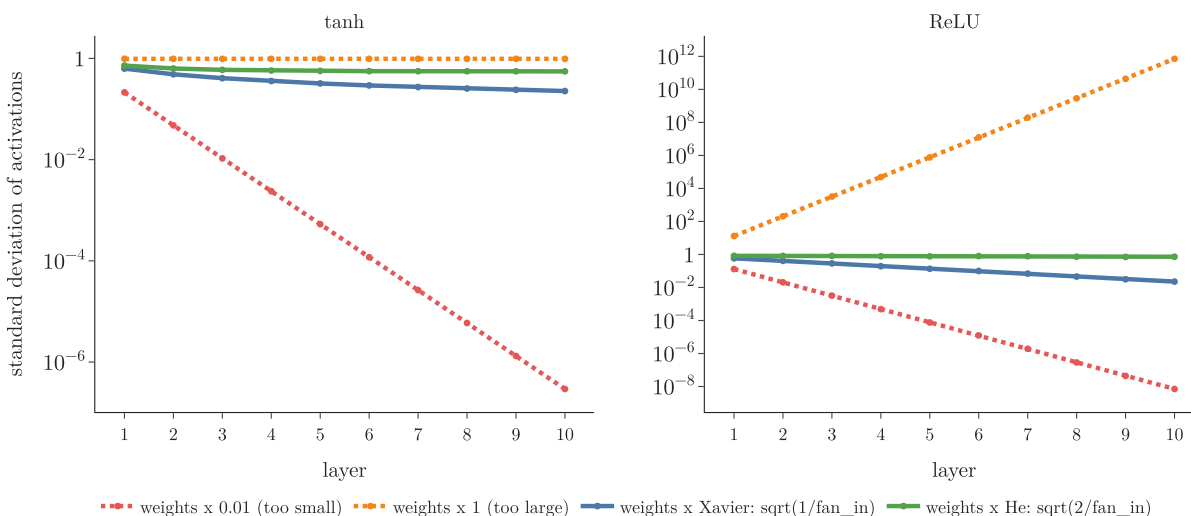


Figure 1: Standard deviation of the activations through 10 layers of 500 nodes. Weights scaled by 0.01 shrink the signal to nothing; scaled by 1, tanh saturates and ReLU explodes. Xavier keeps tanh in range; He keeps ReLU at a steady size

Figure 1 shows the effect. Only the scale of the random weights changes between the lines.

2. Prerequisites

- The [weight initialisation Note](#): the four starts that fail.
- The [measures of dispersion Note](#): variance and standard deviation.
- The [activation functions Note](#): tanh and ReLU.

3. The spread must depend on the number of inputs

Key point: A node adds up n products $w_i x_i$. The more inputs, the larger the sum, so each weight must be smaller. Setting the weights' variance to $1/n$ cancels the effect of n .

3.1 What went wrong with fixed scales

Key point: With 250 inputs, weights of ± 0.01 give a z near 0; weights of ± 3 give a z that can reach ± 250 . A fixed scale cannot suit every layer size.

Take a hidden layer whose nodes each receive 250 inputs. The layer's 250×250 weights were drawn in two ways in the previous Note:

- `np.random.randn(250, 250) * 0.01`: weights around ± 0.01 . Multiplied by inputs around ± 1 and summed, they give a small z , which the activation turns into almost 0.
- `np.random.randn(250, 250) * 1`: weights mostly between -3 and 3 . The sum of 250 such products can be anywhere from about -250 to 250 , and the activation saturates.

The trouble comes from the sum $z = \sum w_i x_i$: its size grows with the number of terms. So no fixed constant works; the scale has to depend on the network's architecture, through the number of inputs to each node.

3.2 Fan-in and fan-out

Key point: Fan-in is the number of inputs coming into a node (the size of the previous layer); fan-out is the number of outputs leaving it.

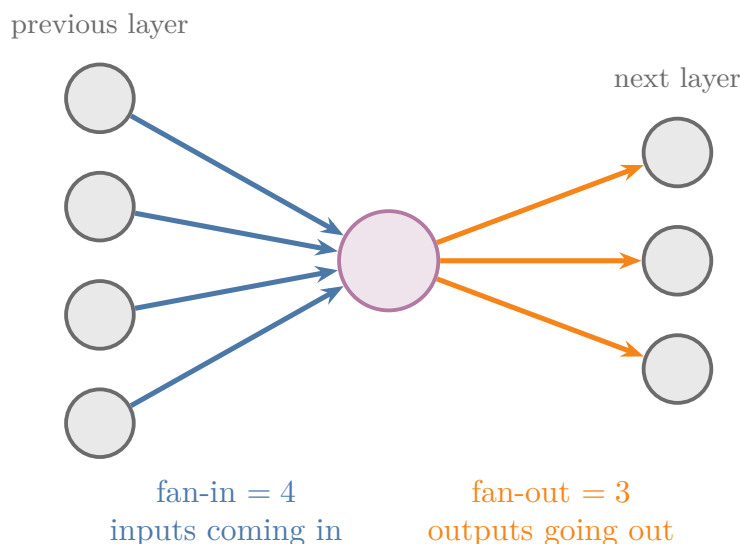


Figure 2: A node with fan-in 4 (inputs from the previous layer) and fan-out 3 (outputs to the next layer)

Two numbers describe where a layer sits (Figure 2):

- **Fan-in:** the number of inputs coming into each node of the layer, which is the number of nodes in the previous layer.
- **Fan-out:** the number of outputs going out of each node, which is the number of nodes in the next layer.

For a Keras `Dense` layer with weight matrix of shape (inputs, nodes), fan-in is the number of rows and fan-out the number of columns.

3.3 Variance $1/\text{fan-in}$

Key point: Many inputs: smaller weights, so the sum stays moderate. Few inputs: larger weights, so the sum is not too small.

The rule is to draw the weights with variance $1/n$, where n is the fan-in. The standard deviation is then $\sqrt{1/n}$, and that is the number we multiply the standard normal numbers by.

1. **In words:** multiply standard normal numbers by $\sqrt{1/\text{fan-in}}$.
2. **Formula:**

$$W = \text{randn} \times \sqrt{\frac{1}{\text{fan-in}}}, \quad \text{Var}(W) = \frac{1}{\text{fan-in}}$$

3. **Example:** with 250 inputs, $\sqrt{1/250} = 0.063$, so the weights are mostly between about -0.19 and 0.19 . With only 2 inputs, $\sqrt{1/2} = 0.71$: fewer inputs, larger weights.

The intuition is a balance. If there are many inputs, each weight is made small so that the sum $\sum w_i x_i$ does not grow too big. If there are few inputs, each weight is made larger so that the sum is not too small. Think of a chain of loudspeakers, each feeding the next: if every speaker turns the sound down, the last one is silent; if every one turns it up, the last one distorts. Each speaker's volume knob must be set so that the sound leaves it as loud as it came in.

Extra: Why exactly $1/n$? For independent w_i and x_i with mean 0, the variance of a product is the product of the variances, $\text{Var}(wx) = E[w^2 x^2] - (E[w x])^2 = E[w^2] E[x^2] - 0 = \text{Var}(w) \text{Var}(x)$, and variances of independent terms add (see the [expected value and variance Note](#) for variance). So

$$\text{Var}(z) = \text{Var}\left(\sum_{i=1}^n w_i x_i\right) = n \text{Var}(w) \text{Var}(x)$$

Choosing $\text{Var}(w) = 1/n$ gives $\text{Var}(z) = \text{Var}(x)$: the signal leaves the layer the same size it came in. The Notebook checks it with 500 inputs of variance 1: $\text{Var}(w) = 0.0001$ gives $\text{Var}(z) = 0.053$ (predicted 0.05), $\text{Var}(w) = 1$ gives 457 (predicted 500), and $\text{Var}(w) = 1/500$ gives 0.93 (predicted 1).

4. Xavier (Glorot) initialisation

Key point: Normal: standard deviation $\sqrt{1/\text{fan-in}}$, or the common variant $\sqrt{2/(\text{fan-in} + \text{fan-out})}$. Uniform: between $\pm\sqrt{6/(\text{fan-in} + \text{fan-out})}$. Use it with tanh and sigmoid.

Xavier initialisation (Glorot initialisation) draws the starting weights with the spread of section 3.3. Glorot and Bengio (2010) designed it for, and tested it on, tanh networks.

4.1 Xavier normal

Key point: Weights from a normal distribution with mean 0 and standard deviation $\sqrt{1/\text{fan-in}}$, or $\sqrt{2/(\text{fan-in} + \text{fan-out})}$.

1. **In words:** standard normal numbers times $\sqrt{1/\text{fan-in}}$. A widely used variant averages fan-in and fan-out.

2. **Formula:**

$$\sigma = \sqrt{\frac{1}{\text{fan-in}}} \quad \text{or} \quad \sigma = \sqrt{\frac{2}{\text{fan-in} + \text{fan-out}}}$$

3. **Example:** a layer with fan-in 250 and fan-out 250:

$$\sqrt{\frac{1}{250}} = 0.063, \quad \sqrt{\frac{2}{250 + 250}} = \sqrt{0.004} = 0.063$$

For equal fan-in and fan-out the two agree. With fan-in 2 and fan-out 10 they differ: 0.71 and 0.41.

The variant with fan-in plus fan-out is the one used more often, and the one Keras calls `glorot_normal`.

Extra: The $\sqrt{1/\text{fan-in}}$ version is often called **LeCun initialisation** (LeCun et al. 1998; Keras: `lecun_normal`), the start SELU needs (see the [ReLU variants Note](#)). Glorot and Bengio (2010) proposed the averaged version, their “normalized initialization”: $1/\text{fan-in}$ keeps the forward signal steady, $1/\text{fan-out}$ keeps the backward gradient steady, and $2/(\text{fan-in} + \text{fan-out})$ is a compromise between the two.

4.2 Xavier uniform

Key point: Weights from a uniform distribution between $-L$ and L , with $L = \sqrt{6/(\text{fan-in} + \text{fan-out})}$.

1. **In words:** every value between $-L$ and L is equally likely.

2. **Formula:**

$$W \sim \text{Uniform}(-L, L), \quad L = \sqrt{\frac{6}{\text{fan-in} + \text{fan-out}}}$$

3. **Example:** fan-in 250, fan-out 250:

$$L = \sqrt{\frac{6}{500}} = \sqrt{0.012} = 0.110$$

so the weights lie between -0.110 and 0.110 .

Extra: The 6 comes from the variance of a uniform distribution: values spread evenly between $-L$ and L have density $1/(2L)$ and mean 0, so their variance is $\int_{-L}^L \frac{x^2}{2L} dx = \frac{L^2}{3}$. Setting $L^2/3 = 2/(\text{fan-in} + \text{fan-out})$ gives $L^2 = 6/(\text{fan-in} + \text{fan-out})$. So Xavier uniform has exactly the same variance as Xavier normal; only the shape differs. Above, $0.110^2/3 = 0.004$, the variance 0.063^2 of the normal version.

5. He initialisation

Key point: Like Xavier, with 2 in place of 1: standard deviation $\sqrt{2/\text{fan-in}}$ (normal) or limit $\sqrt{6/\text{fan-in}}$ (uniform). Use it with ReLU.

He initialisation is Xavier with the variance doubled. He et al. (2015) designed it for ReLU: in their experiments a 30-layer ReLU network trained with it, while with Xavier's spread the same network stalled.

5.1 He normal and He uniform

Key point: Normal: $\sigma = \sqrt{2/\text{fan-in}}$. Uniform: $L = \sqrt{6/\text{fan-in}}$.

1. **In words:** standard normal numbers times $\sqrt{2/\text{fan-in}}$; or uniform numbers between $\pm\sqrt{6/\text{fan-in}}$.
2. **Formula:**

$$\sigma = \sqrt{\frac{2}{\text{fan-in}}} \qquad L = \sqrt{\frac{6}{\text{fan-in}}}$$

3. **Example:** fan-in 250:

$$\sigma = \sqrt{\frac{2}{250}} = 0.089, \qquad L = \sqrt{\frac{6}{250}} = 0.155$$

Both have variance $2/250 = 0.008$, since $0.155^2/3 = 0.008$.

Extra: Why the 2? ReLU sets every negative z to 0. If z is symmetric around 0, the positive half keeps its z^2 and the negative half contributes 0, so $E[\text{ReLU}(z)^2] = \frac{1}{2}E[z^2]$: the size (mean square) of the signal halves at every layer. Doubling the weights' variance makes up for exactly that loss (He et al. 2015). Figure 1 shows it: with ReLU, Xavier's spread lets the standard deviation fall from 0.59 to 0.023 over 10 layers, while He's keeps it between 0.83 and 0.73.

5.2 Which one to use

Key point: Tanh or sigmoid: Xavier. ReLU and its variants: He.

Initialisation	Normal: standard deviation	Uniform: limit L	Use with
Xavier (Glorot)	$\sqrt{1/\text{fan-in}}$ or $\sqrt{2/(\text{fan-in} + \text{fan-out})}$	$\sqrt{6/(\text{fan-in} + \text{fan-out})}$	tanh, sigmoid
He	$\sqrt{2/\text{fan-in}}$	$\sqrt{6/\text{fan-in}}$	ReLU and variants

Both rules come from the variance argument of section 3.3 and were tested in the papers that introduced them (Glorot and Bengio 2010; He et al. 2015). For applying them, the four formulas and this pairing are enough.

6. Xavier and He in Keras

Key point: Pass `kernel_initializer="glorot_normal"`, `"glorot_uniform"`, `"he_normal"` or `"he_uniform"` to each Dense layer. The default is `glorot_uniform`, with biases starting at 0.

6.1 Xavier by hand

Key point: The four-layer tanh network that stayed at loss 0.693 with weights of 0.01 now reaches 0.002 and 100% accuracy.

We reuse the network of section 6.3 of the [weight initialisation Note](#): 300 standardised `make_moons` observations (records), each with two features (input variables) and a class as target (the output we predict), four hidden layers of 10 tanh nodes, plain SGD with learning rate 0.1, 100 epochs. The 0.01-scale network, started from weights of $0.01 \times$ standard normal, never moved from a loss of 0.6934.

Now every weight matrix is drawn as standard normal numbers times $\sqrt{1/\text{fan-in}}$: $\sqrt{1/2} = 0.71$ for the first layer, which has 2 inputs, and $\sqrt{1/10} = 0.32$ for the layers after it, which have 10. The biases start at 0.

Python: Xavier normal by hand.

```
new = []
for w in model.get_weights():
    if w.ndim == 2:                # a weight matrix
        fan_in = w.shape[0]        # rows = inputs
        new.append(np.random.randn(*w.shape)
                    * np.sqrt(1 / fan_in))
    else:                          # a bias vector
        new.append(np.zeros_like(w))
```

```
model.set_weights(new)
```

The loss falls from 0.650 to 0.0023, and the training accuracy is 100%. The weights move: one first-layer weight goes from 0.58 to 2.04. Figure 3 (left) shows the boundary following the moons.

6.2 With Keras' initialisers

Key point: One argument per layer. All four initialisers train the network to 100% here.

Python: He normal for a ReLU layer.

```
keras.layers.Dense(10, activation="relu",  
                    kernel_initializer="he_normal")
```

`kernel_initializer` sets how the layer's weight matrix (Keras calls it the kernel) starts. The other options are "he_uniform", "glorot_normal" and "glorot_uniform"; `bias_initializer` does the same for the biases.

The same four-layer network with each initialiser, 100 epochs of plain SGD:

Activation	Initialiser	Loss at start	Loss at end	Accuracy
ReLU	he_normal	0.770	0.0020	100%
ReLU	he_uniform	0.590	0.0012	100%
tanh	glorot_normal	0.714	0.0032	100%
tanh	glorot_uniform	0.564	0.0021	100%

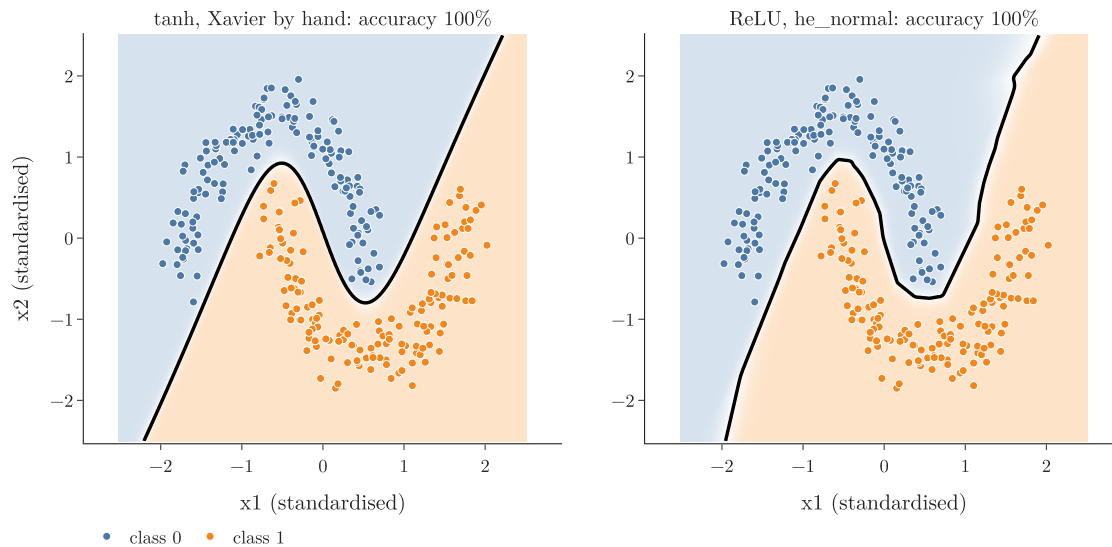


Figure 3: Decision regions after 100 epochs. Left: tanh with Xavier weights set by hand. Right: ReLU with `he_normal`

On such a small problem every good start works. The differences show in deep networks: in He et al. (2015), Xavier and He starts both trained a 22-layer ReLU network, but only He trained a 30-layer one. Trying the combinations on our own data shows which works best there.

The default. If we do not pass `kernel_initializer`, a Dense layer uses `glorot_uniform`, and its biases start at 0 (**zeros**). Keras' default explains why the networks of the earlier Notes trained without our choosing anything.

Extra: Keras' normal versions draw from a **truncated normal** distribution: values more than two standard deviations from 0 are redrawn (Keras documentation). Redrawing alone would shrink the spread, yet the measured standard deviation still matches the formula, so Keras compensates for the cut. The Notebook checks all four on a 1000×100 layer: `glorot_normal` has standard deviation 0.0426 (formula $\sqrt{2/1100} = 0.0426$), and `he_uniform` reaches at most 0.0775 (formula $\sqrt{6/1000} = 0.0775$).

7. Summary

Start	Spread	Signal through 10 layers (Figure 1)
$0.01 \times \text{randn}$	fixed, too small	dies out (tanh and ReLU)
$1 \times \text{randn}$	fixed, too large	tanh saturates; ReLU explodes to 10^{11}
Xavier	variance $1/\text{fan-in}$	tanh stays in range (0.63 to 0.23)
He	variance $2/\text{fan-in}$	ReLU stays steady (0.83 to 0.73)

- Xavier (Glorot): $\sqrt{1/\text{fan-in}}$ or $\sqrt{2/(\text{fan-in} + \text{fan-out})}$; uniform limit $\sqrt{6/(\text{fan-in} + \text{fan-out})}$. For tanh and sigmoid.
- He: $\sqrt{2/\text{fan-in}}$; uniform limit $\sqrt{6/\text{fan-in}}$. For ReLU.
- In Keras: `kernel_initializer=...` per layer; the default is `glorot_uniform`.
- The starting weights must be random, with a spread that depends on the fan-in.

8. Sources

- Glorot, X. and Bengio, Y. (2010). Understanding the difficulty of training deep feedforward neural networks. AISTATS 2010, PMLR 9:249-256 (equation 12: the variance $2/(n_i + n_{i+1})$).
- He, K., Zhang, X., Ren, S. and Sun, J. (2015). Delving Deep into Rectifiers: Surpassing Human-Level Performance on ImageNet Classification. ICCV 2015. arXiv:1502.01852 (section 2.2 and Figures 2 and 3).
- Keras API documentation: layer weight initializers (`GlorotNormal`, `HeNormal`, `TruncatedNormal`), keras.io/api/layers/initializers.
- LeCun, Y., Bottou, L., Orr, G. B. and Müller, K.-R. (1998). Efficient BackProp. In *Neural Networks: Tricks of the Trade*, Springer (equation 16: standard deviation $m^{-1/2}$ for fan-in m).

9. Key terms

Term	Meaning
Fan-in	The number of inputs coming into a node: the size of the previous layer
Fan-out	The number of outputs leaving a node: the size of the next layer
Xavier (Glorot) normal	Starting weights from a normal distribution with standard deviation $\sqrt{1/\text{fan-in}}$ or $\sqrt{2/(\text{fan-in} + \text{fan-out})}$

Term	Meaning
Xavier (Glorot) uniform	Starting weights spread evenly between $\pm\sqrt{6/(\text{fan-in} + \text{fan-out})}$
He normal	Starting weights from a normal distribution with standard deviation $\sqrt{2/\text{fan-in}}$; for ReLU
He uniform	Starting weights spread evenly between $\pm\sqrt{6/\text{fan-in}}$; for ReLU
LeCun initialisation	Normal starting weights with standard deviation $\sqrt{1/\text{fan-in}}$; Keras' <code>lecun_normal</code>
<code>kernel_initializer</code>	The <code>Dense</code> argument that chooses how a layer's weight matrix starts; default <code>glorot_uniform</code>
Truncated normal	A normal distribution with values beyond two standard deviations redrawn